

HIGH STRENGTH PLASTIC

Grey Iron, Alloy Steel,
Stainless Steel,
Copper Alloy,
Nodular Iron

NCTH
NEW CHIEN TE HANG CO., LTD
Importer & Exporter: Foundry Materials
Refractory, Ferro Alloys & Non-Ferro Alloys

We offer a big selection of patching materials for iron casting, steel casting or copper alloy casting.

High Strength Plastic

Product Name		Alumina Base					Silica Base		
		R91	R90	R71	R70	R66S	R81	R80	R66
Component	Al ₂ O ₃	> 90%	> 90%	> 70%	> 70%	> 70%	5%	5%	4%
	SiO ₂	5.50%	5.50%	9%	9%	8%	>90%	>90%	>84%
	Fe ₂ O ₃	1% max	1% max	1% max	1% max	1% max	1% max	1% max	1% max
Density Size		2.5t/m ³ 0-3mm	2.5t/m ³ 0-6mm	2t/m ³ 0-3mm	2t/m ³ 0-6mm	2.2t/m ³ 0-6mm	2t/m ³ 0-3mm	2t/m ³ 0-6mm	2t/m ³ 0-6mm
Max service Temperature (°C)		1800	1800	1700	1700	1600	1550	1550	1550
Application		Grey Iron/ Nodular Iron/Stainless Steel/ Alloy Steel/ Copper Alloy				Grey Iron/ Nodular Iron/Copper Alloy	Grey Iron/ Nodular Iron/ Copper Alloy		Grey Iron/ Nodular Iron/ Copper Alloy
		Ladel/ Furnace Lining/ Runner/Topping				Ladle/Runner	Ladel/ Furnace Lining/ Runner/Topping		Ladle/ Runner

/DIRECTION/

- / Clean and get the construction area wet with water or clay water, and then apply patching materials by hands before smooth the area by trowel
- / After construction is done, air dry for 1-3 hours before drying by heat. Then it can be used.

Note Small cracks are normal situation, because of water evaporating in the process of air dry and heating. Using finer patching material to repair and dry again will be fine.
If the material is too dry to work with, mixing some water, then it can be easy to work again.

NCTH SLAG REMOVER

Copper Alloy Castings,
Cast Iron, Cast Steel,
Stainless Steel,
Alloy Steel



/DESCRIPTION/

- / NCTH Slag Remover (Perlite Sand) is used in the steel and foundry industries for easy removal of slag produced during melting process by coagulating slag or impurities in molten metals to prevent slag inclusion in the casting.
- / NCTH Slag Remover has excellent slag binding performance and significantly fewer toxic smokes or fumes emitted. When applied during ladle filling, a gentle rolling motion is produced which cleanses the metal by coagulating impurities. NCTH Slag Remover can also form an insulating layer to maintain metal temperature when needed.

/DIRECTION/

/ Electric Induction Furnace

Apply NCTH Slag Remover before it reaches pouring temperature can reduce heat radiations and shorten heating-up time. When it reaches pouring temperature, apply appropriate amount of NCTH Slag Remover for easy removal of slag.

/ Ladle

Cover a thin layer of NCTH Slag Remover on the surface of molten metals to prevent slag inclusion in the casting by coagulating impurities when pouring.

/ADVANTAGES/

- / Excellent Slag Binding Performance
- / Few Toxic Smoke or Fumes Emitted
- / Excellent Thermal Insulator
- / Good diffusivity

Note Apply on the surface of molten metals.

Stored in a dry atmosphere or the performance will be affected by moisture.

SLAG X

Copper Alloy Cast,
Nodular Cast Iron,
Gray Cast Iron,
Stainless Steel,
Alloy Steel



/DESCRIPTION/

SLAG X is a kind of additives to soften the slags and purify molten metals to help energy-saving and increase the melting efficiency in electric induction furnace.

IMMEDIATE BENEFITS

- / Soften the Slags
- / Increase the Fluidity of Molten metals
- / Lessen Slags stuck to the Furnace Lining
- / Keep Rider/Feeder clean

SUBSEQUENT BENEFITS

- / Prevent Bridging Effect
- / Extend furnace lining life by 10% and up
- / Lower pouring temperature
- / Reduce casting defects
- / Enhance casting physical property
- / Improve melting efficiency

/DIRECTION/

/ Electric Induction Furnace

Put SLAG X together with the steel scraps and other materials at the bottom of the furnace.

/ Ladle

Put SLAG X at the bottom of the ladle and then put the molten metals in.

/ Addition amount

0.03%-0.08% of molten metals. 0.05% is suggested for the initial use. Observe the condition of the furnace lining after each run and then modify the amount accordingly.

Note Do not apply directly on the surface of the molten metals.
Store in a dry atmosphere otherwise the effectiveness will be affected due to moisture.
Pay attention to changes of furnace lining as operation condition vary.

EXOTHERMIC ANTI-PIPING COMPOUND

Nodular (Ductile) Iron,
Stainless Steel,
Alloy Steel,
Grey Iron



/DESCRIPTION/

- / NCTH Exothermic Anti-piping Compound is used in foundries after pouring, to maintain the molten metals upper temperature nearly the same as the lower layer, and prevent piping defect layer from happening .
- / Exothermic Anti-Piping Compound keeps the molten metals temperature around the riser/feeder by burning itself to form an insulation layer to assure the temperature dropping more evenly during the molten metals solidifying process.
The shrinkage status of molten metals around riser/feeder change from the V shape to the U shape.

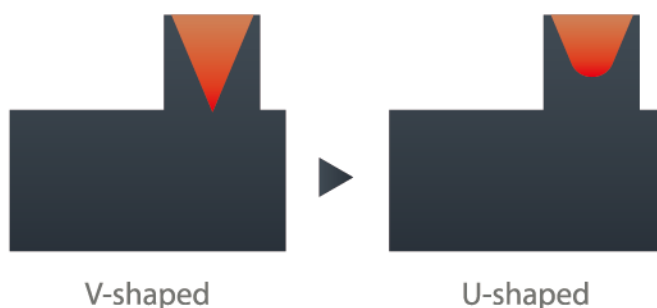
/DIRECTION/

- / After pouring, apply the compound on the surface of molten metals through sprue and riser/feeder head. The compound will react with molten metals and start burning to supply heat.

/ADVANTAGES/

- / Slow Down Molten Metals Solidifying
- / Reduce Heat Loss
- / Improved Feeding Capacity
- / Decreased Piping Defect
- / Few Toxic Smoke or Fume Emitted

/SHRINKAGE COMPARISON/



Note Store the compounds in a dry, cool and shady area, and away from any heat, flame or anything that sparks.

Plus⁺ Low Ash Sea Coal



/Function/

- / **Form Gas Cushion:** To block molten metals in direct contact with the surface of sand mold and prevent burn-on and burn-in from happening
- / **Offer Reducing Atmosphere:** To redox oxidized layer of molten metals surface and prevent it from being further oxidized and burn-on defects due to the reaction between castings and sand on the mold surface.
- / **Increase the collapsibility of sand mold:** To make the mold surface smooth and demolding easier, and to reduce burned sand and penetration defects



- ▲ Through a controlled and strict manufacturing process, it assures the product liability.

SUPER BOND+

Resistant Cast Iron,
Ductile Cast Iron,
Stainless Steel,
Alloy Steel

/PRODUCT CHARACTERISTICS/

- / Increase green sand strength more than 20%
- / Increase Mold Surface Stability more than 25% and improve stability of green sand
- / Increase the recyclable amount of used-sand and decrease the hottest sand stuck amount
- / Increase sand mold detailed parts and corners fulfilled and decrease sand mold partially failure rate
- / Lustrous Carbon provides reducing atmosphere and better air permeability in order to reduce the castings failure rate and have better precision in size and appearance.
- / Low water and volatile content ; low adding amount.
- / Better collapsibility to shorten the post processing time.



Addition amount 0.05-0.08% (Ex: 1000kg
recycle of sand, addition amount 0.5-0.8kg)

Plus* BentoBond



/DESCRIPTION/

Based on artificially sodium-activated bentonite, we have combined a unique non-starch, non-dextrin material carefully formulated for high-temperature conditions.

This formula enhances dry compression strength by 7% compared to natural sodium bentonite under severe temperature conditions (above 45°C), while also demonstrating excellent mulling efficiency. This allows you to achieve ideal green strength within effective mixing times.

Bentobond's moderate swelling ensures excellent moldability, effectively reducing mold defects and improving mold quality, with a 7% increase in permeability compared to natural sodium bentonite.

With our product, you can achieve higher efficiency and more stable green sand quality.

/DIRECTION/

Add the amount of existing clay and adjust according to process requirements.

/ADVANTAGES/

- Improved clay addition in high sand temperature conditions
- Excellent moldability and blowing properties
- Increased mulling efficiency
- Strengthened dry compression strength
- Enhanced mold permeability
- Reduced mold defects

Plus* 
Experimental
Data

	真佳PLUS+ BentoBond	Natural Sodium Bentonite
Moisture(%)	2.96	3.00
Compactability(%)	39	40
Green Compressive Strength (kg/cm2)	2.09	2.08
Permeability	88  7% UP	82
Surface Stability Index(%)	91.69	91.75
Wet Tensile Strength (kg/cm2)	0.0329	0.0351
Dry Compressive Strength (kg/cm2)	3.26  7% UP	3.05